COUNTRY PROFILES

Uranium in Namibia

UPDATED WEDNESDAY, 21 JANUARY 2026

- Namibia has significant uranium mines capable of providing 10% of world mining output.
- Its first commercial uranium mine began operating in 1976.
- There is strong government support for expanding uranium mining and some interest in using nuclear power.

Uranium was discovered in the Namib Desert in 1928, but was not until intensive exploration got under way in the late 1950s that

Privacy - Terms

much interest was shown in Rössing. Rio Tinto discovered numerous uranium occurrences and in 1966 took the rights over the low-grade Rössing deposit, 65 km inland from Swakopmund.

Two other significant deposits found in early exploration were Trekkopje, a calcrete deposit 80km northeast of Swakopmund and near Rössing, and Langer Heinrich, a calcrete deposit discovered in 1973 by Gencor, 80 km inland from Walvis Bay and 50 km southeast of Rössing.

In April 2011 the Namibian government announced that its state-owned mineral exploration company, Epangelo Mining Ltd, would have exclusive control over new strategic minerals developments, including uranium. However, this does not apply retrospectively or amount to nationalization of existing mines or leases. Paladin and Kalahari have both expressed confidence that their assets are not at risk of expropriation, and in April 2012 Epangelo agreed to buy a share of the Etango project and pay a corresponding share of development costs. A task force was formed in May 2010 by Epangelo and Russia's ARMZ, which briefly seemed to threaten existing developments, but the government gave strong reassurance then, borne out since. New exploration licences will be granted only to Epangelo, and others interested will need to negotiate farm-ins with it, to become joint-venture partners.

Over 2007-2010 a strategic environmental assessment was undertaken over the whole uranium province inland from Swakopmund and Walvis Bay. This addressed the whole region and all the projects. It resulted in the Strategic Environmental Management Plan implemented in 2011 by the government and individual project companies.

	Deposit type	Resources	
		Measured & indicated	Inferred
Rössing	hard rock	23,810 probable reserves	0
Langer Heinrich	palaeochannel	34,785 tU in 0.042% ore	3454 tU in 0.04% ore
Trekkopje	palaeochannel	0	26,000 tU in 0.01% ore
Husab	hard rock	143,740 tU in 0.048% ore	40,130 tU in 0.04% ore
Norasa	hard rock	39,700 tU in 0.0167% ore	8500 tU in 0.014% ore

	Deposit type	Resources	
		Measured & indicated	Inferred
Etango	hard rock	63,000 tU in 0.016% ore	24,000 tU in 0.016% ore
Marenica	palaeochannel & hard rock	22,040 tU in 0.01% ore	22,039 tU in 0.01% ore
Omahola	hard rock	29,160 tU in 0.016% ore	19,140 tU in 0.016% ore
Tubas-TRS	aeolian	0	10,900 tU in 0.0125% ore
Tumas	palaeochannel	40,200 tU in 0.028% ore	6300 tU in 0.019% ore

Namibia Uranium Production – tonnes U

		2010	2011	2012	2013	2014	2015	2016	2017	2018
Rössing	CNUC	3083	2641	2289	2043	1308	1057	1569	1790	2102
Langer Heinrich	Paladin	1419	1437	1960	2098	1947	1937	1893	1294	394
Husab	Swakop						0	192	1140	3028
Trekkopje	Orano	0	0	251	186	0	0	0	0	0
Total			4078	4500	4327	3255	2994	3654	4224	5524

Both China and India are interested in acquiring uranium from Namibia. Chinese companies have taken major equity positions, notably with Husab, but also Langer Heinrich. In November 2018, Rio Tinto announced it had agreed to sell its majority stake in Rössing Uranium Limited to China National Uranium Corporation Limited (CNUC). India is discussing how to open the way to be able to buy Namibian uranium, at present ruled out by Namibia's non-proliferation commitments – see final section below.

The Namibian Uranium Association and its Namibian Uranium Institute provide an industry overview.

Water supply

One of the aspects of the Strategic Environmental Management Plan is water supply. Since 2010 water has been supplied to

Trekkopje from a coastal reverse osmosis (RO) desalination plant 20 km north of Swakopmund in the Erongo region with about 20 million m^3/yr output and requiring 16 MWe from the grid. Some of this water, up to 10 million m^3 per year, is available to mines other than Orano's. The plant current supplies 12 m^3/yr to Namibian Water Corp (NamWater) for Rössing (reported to require about 3 million m^3/yr) and Husab (reported to require about 2 million m^3/yr). Langer Heinrich was also being supplied earlier. However, the commercial arrangements for maintaining and operating the plant are contentious, and lead to costs to these companies of at least $4 per cubic metre.

The plant is owned by Erongo Desalination Company which was to be a joint venture of Orano and a local company, United Africa Group, but the latter is no longer involved. It is operated by Aveng Water Treatment Namibia. Orano Resources Namibia was in discussions to share its stake to Nambia Water, but this did not proceed. In 2016 it put a price of N$ 3 billion ($216 million) on the plant, which is capable of expansion to 26 million m^3/yr. The government declined the offer, and said it would proceed with building a new plant in a public-private partnership, reviving an earlier proposal for a 60 million m^3/yr capacity plant adjacent, costing $145 million.

The Ministry of Agriculture, Water & Forestry conducted a feasibility study for the plant with German state-owned KfW Development Bank. It was to be built and operated by “a competent private firm, and Namibian Water Corporation, or alternative consuming entity, shall have a water purchase relationship with the desalination facility." But this did not proceed.

In late 2025, Namibia Water finalized a joint venture with Swakop Uranium, a subsidiary of China General Nuclear (CGN), for a new 20 million m^3/yr desalination plant.

The Ministry of Agriculture, Water & Forestry conducted a feasibility study for the plant with German state-owned KfW Development Bank. It would be built and operated by “a competent private firm, and Namibian Water Corporation, or alternative consuming entity, shall have a water purchase relationship with the desalination facility." A tender was envisaged in 2019.

Earlier in July 2014 Rössing Uranium said that it had been unable to reach agreement on purchasing water long-term from Erongo at a satisfactory price, so would set up its own 3 million m^3/yr (8200 m^3/d) RO plant near Swakopmund. However, the Ministry of Environment & Tourism has rejected this proposal.

Sources such as the Omaruru delta and Kuseb River aquifers dwindled in 2013 during Namibia's worst drought for 30 years, leaving the coastal towns of Swakopmund and Walvis Bay short of water, and in August 2016 the President declared a state of emergency since the drought then was unparalleled. The Erongo plant supplies some water to Swakopmund.

Rössing

Rössing Uranium Ltd was formed in 1970 (68.6% China National Uranium Corporation Limited – CNUC, 15% Iranian Foreign Investment Co., 10% Industrial Development Corporation of South Africa, 3% Namibian government). The company has mined the deposit from 1976 as a large-scale open pit in very hard rock. Rössing has nominal capacity of 4000 tU/yr. Annual production is tabulated above.

In March 2020, Rössing suspended normal mining operations in response to the coronavirus pandemic. Minimal mining operations and essential mine and plant maintenance continued, and already-mined ore was used to minimize the impact on production. Operations subsequently resumed. Work on the Phase 4 pushback began in December 2023, with production scheduled to commence in 2026. A 15 MWe photovoltaic solar power plant is under construction to supply the majority of Rössing's power needed during the day.

Rössing's reserves at the end of 2018 were 23,810 tU probable, at 0.033% U in ore and no resource figures were published. Its uranium is sold to power utilities in Central Europe, North America and Southeast Asia including China.

After three years evaluation it was decided in December 2005 to invest US$ 112 million to further develop the Rössing mine, extend its life to 2016 and increase the output to 3400 tU/yr. In 2007 Rio Tinto proposed a further expansion to 3800 tU/yr from 2012 and extending mine life. The first phase extended mining in 2008 to a new small orebody, introduced radiometric ore sorting to beneficiate material from stockpiled coarse ore, and commissioned a new 1200 t/day sulfur-burning acid plant also producing 9.5 MWe net of electricity. Phase 2 defined in 2008 included heap leaching of low-grade ore and development of other small satellite orebodies with different mineralization and hence needing a new treatment plant for them. For Rössing, 2007 was a year of consolidation, preparing for increased production. Unit costs

therefore rose to US$ 38 per pound ($99/kgU) from $22 ($57/kgU) in 2006.

In 2013 resources for the Z20 orebody, south of the main pit, were quoted, with 46,274 tU at higher grades than the main orebody, and in 2014 development was said to be feasible when the uranium price returned to $60/lb. A pre-feasibility study was undertaken in 2015-16 on the development of Phase 4, with potential to extend the life-of-mine from 2025 to 2031 to recover an additional 18,000 tonnes U_3O_8. This proved uneconomic and the project was deferred to allow the market to recover to the project break-even uranium price of $60/lb.

In November 2018, Rio Tinto agreed to sell its 68.6% stake in Rössing to CNUC. The deal included an initial cash payment of $6.5 million, paid at completion, and a contingent payment of up to $100 million following completion of the sale. The sale required approval by the Namibia Competition Commission (NaCC). The sale was completed in July 2019.

Following the 2019 change of ownership the life-of-mine extension feasibility study was completed in 2023, with board approval in February 2023 for the Phase 4 pushback.

The tailings storage facility is being expanded to accommodate ten years of additional tailings. Additional capacity would be required for life-of-mine expansion to 2035.

In February 2023 the company's board approved extending the operating life of the mine until 2036. A 13-year contract was signed with Beifang Mining for contract mining services from 2024 to 2036.

Earlier in 2011 the Namibian government initiated negotiations with Iran regarding holding in trust the Iranian 15% share in Rössing while UN sanctions on Iran apply, or Epangelo buying that share.

Langer Heinrich

Paladin's Langer Heinrich is 50 km south-southeast of Rössing, in the Namib Park, and 80 km from the coast. It was bought by Paladin Resources (now Paladin Energy) in 2002. The open pit mine commenced operation late in 2006 with 1000 tU/yr capacity. The ore occurs over 15 km in a palaeochannel system, some 50m deep. Some vanadium is present in the carnotite mineral. There is a conventional hard rock mill with an alkaline leaching circuit.

Annual production by calendar year is tabulated above. Stage 3 development boosted production capacity to 2000 tU/yr from late 2011 at a cost of US$ 100 million. In 2015-16 the C1 cash costs were $25.38/lb, with sales at $37.75/lb U_3O_8. Recovery in late 2014 was 84.5%. A heap leach to produce about 400 tU/yr from low-grade ore by mid-2014 was proposed for stage 4, but did not proceed.

In 2017 mining was curtailed and medium-grade stockpiles reduced, leaving some higher-grade ore for when uranium prices improve. C1 cash costs were reduced to $19.9/lb, due to less material being moved. Stockpiled medium-grade ore was largely depleted in April 2018, and in May Paladin announced it would suspend operations at the mine pending improvement in prices. It went into care and maintenance in August.

Production resumed in March 2024 following improved uranium market conditions.

Reserves are 32,300 tU at 0.038%U, with 0.021%U cut-off (JORC and NI 43-101 compliant) including stockpiles. Most of the 2008 inferred resources have been upgraded with infill drilling. A further 12,000 tU was intended to end up in low-grade stockpiles at the end of mining, with a view to possible recovery then. In mid-2017 a trial program was commenced with Marenica's U-pgrade technology on ore samples, with a view to possible beneficiation of the low-grade ore as at Tumas (see below). Both companies reported this as being successful on the low-grade ore, with 35-fold grade increase and 80% recovery so that one tonne of ore resulted in only 25kg of mill feed.

Earlier in January 2014 China National Nuclear Corporation's subsidiary CNNC Overseas Uranium Holding Limited (CNNC Overseas) bought a 25% joint venture equity stake in the mine for $190 million, entitling it to that share of output. In mid-2016 Paladin agreed to sell another 24% of the equity for $175 million, also to CNNC, but this did not proceed. In March 2017 CNNC Overseas asked Paladin to begin determining "fair market value" of the operation so that it might possibly exercise an option to acquire the 75% balance of it. In August 2017 CNNC Overseas decided not to proceed with acquiring any further equity.

Husab

Swakop Uranium started development of the mine in February 2013, and production commenced at the end of 2016, with 192 tU

in 2016, and ramped up to approximately 5300 tU/yr against a design capacity of 6000 tU/yr, with most of this being supplied to China and up to 20% being marketed internationally by CGN Global Uranium Ltd, incorporated in the UK.

In March 2020, mining operations were suspended in response to the Covid-19 pandemic. Operations subsequently resumed. A heap leach pilot plant became operational in July 2024.

Swakop Uranium is now 90% owned by Taurus Minerals and 10% by Epangelo Mining Co. Taurus is 60% owned by China's CGN-Uranium Resources Co (CGN-URC) and 40% by China-Africa Development Fund, set up by China Development Bank in 2007. In July 2014 CGN Global Uranium Ltd (CGU) was incorporated in the UK to sell Husab uranium on the world market.

The main part of the Husab project, formerly referred to as the Rössing South orebody, is about 5 kilometres south of the Rössing mine and 45 km northeast of Walvis Bay port. A definitive feasibility study demonstrating the technical and economic viability of mining Zones 1&2 was completed by Perth-based Extract Resources on the basis of measured resource for them of 32,000 tU averaging 0.043%U, and indicated resources of 105,500 tU at 0.037%U (JORC and NI 43-101- compliant). Inferred resources in Zones 1 to 5 are 50,000 tU averaging 0.029%U. This comes to 188,000 tU averaging about 0.035%U proven to June 2011, all with 100 ppm cut-off and still open along strike and dip. It is evidently the highest grade granite-hosted uranium deposit in Namibia, and it is an extension of the Rössing stratigraphy. Only 10km of 15km strike on the company's lease – contiguous with Rössing – had then been drilled. It lies under a shallow (50m) alluvial sand cover. Zone 1 pit will be 1.3 x 2.5 km and Zone 2 pit will be 1.0 x 1.5 km.

The company's Husab uranium project includes the Ida Dome 20 km south, with 9600 tU inferred resource at 0.02% (contiguous with Reptile's Ongalo).

The definitive feasibility study involved the project's zones 1&2, and showed a production cost of US$ 32/lb U_3O_8 including royalties, marketing and transport, with capital cost of $1.66 billion. The study envisages mining of 15 million tonnes of ore per year from two separate open pits to feed a processing plant producing 5770 tU per year. The Ministry of Mines and Energy approved the mining licence in November 2011, and in January and July 2011 the Ministry of Environment and Tourism gave environmental approval for the mine and related works to subsidiary Swakop Uranium, which holds some of the tenements. Water supply and labour

agreements were signed in April 2014. Mine development commenced in May 2014. A 1500 t/day sulfuric acid plant is built and additional acid may be imported. Late in 2016, water supply was a problem, and 140 million tonnes of overburden had been moved. The EPC contract was the largest for any African mine. First production was at the end of 2016.

Corporate history: The two-stage takeover by Taurus/CGN-URC was complex. Kalahari Minerals PLC owned 42.74% of Extract Resources. In March 2010 Itochu Corp* bought a 14.94% stake in Kalahari for $92 million, and in May Hong Kong-listed APAC Resources bought 7.1% of Kalahari for $44.6 million. In March 2011 China Guangdong Nuclear Power UraniumResources Co (CGN-URC) notified a possible $1.22 billion cash offer for the whole of Kalahari Minerals PLC, but this was withdrawn in May. It was then renewed, and in December a GBP 632 million (US$ 984 million) bid by Taurus Minerals, a Hong Kong company 60% owned by CGN-URC and 40% by the China-Africa Development Fund, was unanimously recommended by Kalahari directors. It then acquired some 90% of Kalahari, including Rio Tinto's 11.5% and Itochu's 14.9% stake, requiring it to make a $2.2 billion downstream cash offer for Extract shares. It did this, Extract directors recommended acceptance, and the takeover was completed in April 2012. CGN-URC became CGNPC-Nuclear Fuel Co (CGNPC-NFC) in February 2012.

* through Nippon Uranium Resources (Australia) Pty Ltd., Itochu's wholly-owned Australian subsidiary. Also, Itochu had a 10.3% direct stake in Extract.

In November 2012 the Namibian government's Epangelo Mining Company agreed to buy a 10% interest in Swakop Uranium for $213 million, funded by the vendor and to be repaid from dividends**. While CGN-URC is majority shareholder, some of the production will be sold on world markets.

** Epangelo had for some years been a potential partner in the project. A joint task force announced in May 2010 comprising Russia's ARMZ and Epangelo Mining Ltd with $1 billion funding was reported to be "aimed at Rössing South", but Extract received assurance from the government that its leases were secure, and Namibia's international reputation supported this.

Extract Resources had earlier rejected overtures from Rössing Uranium for joint development, using the Rössing treatment plant, though in February 2011 Extract confirmed that it was holding discussions with Rio Tinto about combining its Husab Uranium

Project with Rio's adjacent Rössing Uranium mine (68.6% owned), "with a view to capturing the significant potential synergies that could be generated from a joint development of the two projects". These discussions were inconclusive. Rio Tinto had direct equity in Extract of 14.22%, and 11.52% of Kalahari (formerly 19% overall in Husab) but sold both holdings to Taurus.

Reptile

Australia's Deep Yellow, through wholly-owned subsidiary Reptile Mineral Resources (formerly Reptile Uranium Namibia), is investigating the Reptile project (formerly Omahola project). It comprises both Omahola alaskites and paleochannel and sands south of these, especially Tumas, now central to the Reptile project.

Omahola Basement/Alaskite includes the high-grade Inca primary uraniferous magnetite deposit at about 200 metres depth, the Ongolo Alaskite 10 km away, and MS7 Alaskite in between and possibly connected to it. An acid leach mill near the Ongolo Alaskite deposit was envisaged treating ore from Ongolo and MS7 Alaskite deposits and the Inca uraniferous magnetite deposit, all possibly mined by shallow open pit (to 200 m).

Inca has 2800 tU indicated and 2400 tU inferred resources at about 0.04%U. It is about 10 km south of Etango and 35 km in from the coast. Some 12km northeast of Inca, the Ongolo Alaskite deposit was discovered in 2010, and has 2600 tU measured, 3000 tU indicated and 4000 tU inferred resources at 0.032%U grade and a strike length of up to 2km (contiguous with Extract's Ida Dome). In between is MS7 with 1660 tU measured, 370 tU indicated and 500 tU inferred resource. It is 600m along strike and 400m wide.

All resources (Jan 2013) are JORC-compliant. In 2021 the company reported resources of 43,800 tU grading 0.016%U at 100 ppm cut-off for the whole greater Omahola project comprising Inca, Ongolo and MS7.

In November 2011 the company submitted an environmental assessment report for Inca envisaging an open pit mine producing up to 2.5 Mt/yr of uranium and iron-bearing ore which could result in production of up to 960 tU/yr, depending on project economics. The company has applied for a mining licence. Environmental assessment of Ongolo-MS7 was done in 2012. A mill situated close to Ongolo was envisaged. However, the company is now focusing on the paleochannel deposits.

Test work on Ongolo alaskite ore reported in April 2013 showed promising results from heap leaching. In June 2014 the results were confirmed, with potential uranium recoveries of 80% after seven days' leaching and acid consumption of 12.4 kg per tonne. This means that the cut-off grade could be 100 ppm and hence average grade of the ore would be reduced to 0.0191%, with large increase in recoverable uranium to 48,000 tU.

Tumas & Tubas paleochannel/calcrete

The paleochannel/calcrete-type uranium resource base includes the Tumas 1, 1E, 2&3 and Tubas deposits and the overlying Red Sands resource. The major paleochannel has several zones stretching over about 38km, south and southeast of the Omahola alaskites. In February 2021 the company announced a maiden ore reserve of 11,900 tU of probable reserves for Tumas. In September 2021 as part of a definitive feasibility study it updated JORC figures for all Tumas deposits to 38,000 tU indicated resources. In October it announced probable reserves of 26,300 tU. In February 2023 Deep Yellow released an updated definitive feasibility study indicating a production capacity at the Tumas plant of 1,385 tU per year, with throughput believed to be at 4.15 million tonnes per year. A mining licence was issued by Namibia's Ministry of Mines and Energy in late 2023.

The Tubas Sand deposit is a westward extension of Tumas 3 and 10 km south of the Omahola alaskites. It comprises the shallow aeolian Tubas Red Sand (TRS) deposit immediately south of the Tubas palaeochannel. It has inferred resources of 4900 tU as carnotite, which can be readily beneficiated to 0.05% using German hydrocyclone technology. There are also calcrete ores in the Tubas palaeochannel. Development would be with Tumas.

Aussinanis, Yellow Dune

Reptile's Aussinanis project, also a palaeochannel deposit, near the coast about 60km south of the others. It has 6976 tU indicated and inferred resources at about 0.02%U, and the hydrocyclone technology tested on TRS has potential application here. In January 2013 Deep Yellow agreed with Namibia's Epangelo to transfer its Aussinanis and Ripnes projects into a new company, **Yellow Dune Uranium Resources** Ltd. Epangelo acquired a 5% stake in Yellow Dune to fund test work and confirm that the Aussinanis deposit can

be upgraded by beneficiation. Reptile holds 85% and Oponona Investments 10%. If the test work at Aussinanis is successful, Epangelo will become the operator of the joint venture and would earn up to 70% in Yellow Dune by funding the project through to a bankable feasibility study. Reptile's holding would then shrink to 20%.

Nova

Nova Energy is a joint venture with Reptile Uranium (65% share) and holding an exclusive prospecting licence (EPL 3670) contiguous on the south with its tenement holding Inca and Tubas. It and EPL 3699 near Inca and Etango are prospective both for alaskite and palaeochannel calcrete. Nova Africa, a subsidiary of Toro Energy, holds 25% and Sixzone Investments (a Namibian group) holds 10%. In March 2017 Deep Yellow agreed to a farm-in by Japan Oil, Gas and Metals National Corporation (JOGMEC), which can earn a 39.5% share of Nova with expenditure of A$ 4.5 million in four years. Reptile will continue to manage the project.

Paladin held 15% of Deep Yellow until December 2016.

Trekkopje

Orano's Trekkopje is about 80 km northeast of Swakopmund, and 35 km north of Rössing. In 2007 UraMin Inc announced an upgrade of uranium resources at this project, comprising two adjacent palaeochannel deposits (Klein Trekkopje being the main one) over an area about 16 km by 1 to 3 km. The company was then taken over by Areva (now Orano) to become Areva Resources Southern Africa, with subsidiary Areva Resources Namibia developing the mine (Orano 100%).

The $1 billion project has a shallow open-pit mine and plans a sodium carbonate/bicarbonate heap leach process. About 80 percent of the ore is less than 15 metres deep, but is very low grade – 0.012-0.015%. Since 2010 water has been supplied from a coastal desalination plant set up by Areva with about 55,000 m^3/day (20 million m^3/yr) output. Some of this water is available to other mines. The plant was jointly owned by Areva and a local company, United Africa Group, and was to remain in operation despite the mine's mothballing.

A substantial conversion of 'inferred' resources to reserves occurred as a result of drilling in 2006 and 2007, taking the measured and indicated resource category to 42,000 tU in the main deposit. Areva quoted 45,600 tU resources in 2008, but then revised this to 26,000 tU in 2011 at lower grades, as it announced a €1.8 billion write-down of its investment. Over 9000 tonnes of vanadium pentoxide by-product was envisaged. The mine was intended to produce 3200 tU/yr from 2013.

A mining licence was granted in June 2008, first concentrate from the pilot phase was produced in January 2011 but development stalled in October 2011 due to low uranium prices. The second stage pilot operation had been commissioned in mid-2010, and the main ore stacking for on-off alkaline leach operation commenced early in 2012. However, the project was put onto a care and maintenance basis in October 2012: "considering both the continued decrease of uranium prices coupled with the investments yet to be made on site, Areva has no other option than to postpone the launch of the Trekkopje mine." Production in 2012-13 was 437 tU, “demonstrating the feasibility of the technical solutions adopted and confirming the production cost targets.”

Norasa (Valencia + Namibplaas)

Forsys Metals Corp. of Toronto was developing the **Valencia** uranium project along strike from Rössing and 25 km northeast of it, with geology (alaskite) similar to Rössing. Environmental approval for an open pit mine was granted in June 2008 and a 25-year mining licence was granted in August 2008 to Valencia Uranium P/L (a wholly owned subsidiary of Forsys). Further drilling from 2010 has been at the **Namibplaas** area, 7 km northeast, with similar geology and which Forsys now fully owns through Dunefield Mining P/L.

Consolidated measured and indicated resources figures (October 2013, NI 43-101 compliant) are 39,700 tonnes U at 0.0167% U with 100 ppm cut-off at Valencia and 160 ppm at Namibplaas. This includes reserves of 35,000 tU at 0.017% U. Inferred resources are 8500 tU at same grade. The Valencia open pit will be 1600x1000 metres and 450 m deep, with a small satellite pit adjacent. The Namibplaas pit will be 1900x500 m and 210 m deep.

The Norasa project is a proposed development involving both deposits. A March 2015 feasibility study estimated annual production of about 2000 tU over a 15-year mine life. Costs are estimated at $32.96 per lb U_3O_8 over the first five years of

production and $34.72 per lb U_3O_8 over the mine life. Test work has improved uranium recovery from 85% to 91% by using hydrogen peroxide rather than pyrolusite (MnO_2) as the leach oxidant. Radiometric sorting is planned.

Etango

In October 2010 Perth-based Bannerman Resources Ltd, now Bannerman Energy Ltd, announced measured and indicated resources of 57,330 tU at about 0.019% and inferred resources of 24,600 tU at slightly lower grade, both with 100 ppm cut-off (JORC & NI 43-101 compliant) for the Etango project 30 km southwest of Rössing and 35 km east of Swakopmund. This was formerly known as Goanikontes. The inferred resources are mostly at the adjacent Ondjamba and hyena orebodies. The alaskite ore is very similar to that at Rössing, up to 400m deep, but with two thirds of the resource less than 200 m deep, and open pit mining is envisaged. From trials in a pilot plant using four 2m square columns, each 7m high, heap leaching appears to be the most cost-effective recovery method. Environmental approval for development of the project was received in 2010, and that for infrastructure in mid-2011. A mineral deposit retention licence was issued in October 2017, enabling mining.

A definitive feasibility study was completed by AMEC Minproc in March 2012, confirming the viability of the project on the basis of measured and indicated resources only, and putting its cost estimate at $870 million. Some 80% of measured and indicated resources were converted to proven and probable reserves of 46,000 tU at 0.0165%U, supporting minimum open pit life of 16 years. Production at 2700-3500 tU/yr is now envisaged over the first five years, then 2300-3100 tU/yr, with production costs of $41/lb U_3O_8 over the first five years and $46 longer-term. In September 2014 the company awarded contracts to construct and operate a heap leach demonstration plant, which was commissioned in March 2015. In July and then November 2015 the company said that this supported the assumptions in the definitive feasibility study, with over 90% U recovery and only 15 kg/t acid consumption. It thus demonstrated the viability of the operation for potential investors and financiers. A further three phases of leach and SX testing ran successfully to mid-2016, and operation of the demonstration plant was extended. A project schedule drawn up as part of the definitive feasibility study envisages an engineering and construction period of about 30 months from project approval to plant commissioning. A further definitive feasibility study based on

results from the demonstration plant was commissioned from AMEC Foster Wheeler in mid-2017 and completed in 2022.

In August 2020 the company announced its Etango-8 scoping study, reducing throughput to 40% of that previously envisaged in the definitive feasibility study and focused on shallow, higher-grade ore. A preliminary feasibility study in mid-2021 then confirmed capital cost reduced to $274 million, and final cash operating cost (apart from royalties) to $39.5/lb. Production would be 1350 tU per year over 15 years. The option of subsequent expansion to the levels of the definitive feasibility study remains. Construction of early works commenced in 2024.

Bannerman originally held 80% of the Etango project and in December 2015 acquired the balance of it. It has been seeking a development partner. In July 2011 China's Sichuan Hanlong group made a conditional A$144 million takeover offer for Bannerman, but this did not proceed. In April 2012 state-owned Epangelo Mining Ltd agreed to buy a 5% stake in the project for about $3.9 million, with an option to buy a further 5% upon commitment to mine development. However the deal was called off when the companies could not agree on terms. Funding was formerly from a major shareholder, Resource Capital Funds, but at the end of 2015 the company was debt-free and looking forward to project funding. In July 2017 the One Economy Foundation in Namibia became a 5% loan-carried shareholder in the project, as a "key pillar" in Bannerman's corporate social responsibility. The Windhoek-based organization works to bridge the divide between the country's prosperous and knowledge-based urban 'first economy' and the township or rural-based 'second economy' which lacks access to the networks, technology and capital required to succeed in the first economy.

Marenica

In July 2008 West Australian Metals, re-named Marenica Energy, now re-named again Elevate Uranium, announced a modest JORC-compliant inferred resource in the Marenica palaeochannel deposit 30 km north of Areva's (now Orano's) Trekkopje and similar to it. In December 2011 the company revised this to indicated resources of 2500 tU at 0.01%U, and an inferred resources of 19,600 tU in 0.008%U ore, mostly palaeochannel but with some granite-alaskite basement rock down to 60 metres. These figures were current in mid-2021. There is good potential to significantly upgrade the uranium plant feed grades by sizing and gravity separation,

achieving beneficiation from 0.01% to 0.55% U_3O_8, and increasing the leach feed grade accordingly. The uranium occurs as carnotite of a particular particle size, significantly denser than the host gangue, mainly calcite. This upgrade potential effectively dismisses heap leaching as a process option with the preferred process now being acid tank leaching of the beneficiation product. (Earlier, the company was considering a $260 million heap leach operation with production of 1350 tU/yr over 13 years at $82/lb.)

Marenica/Elevate also holds extensive exploration tenements south of Paladin's Langer Heinrich, and east/upstream of Deep Yellow/Reptile's Tumas. Adjacent to Tumas is its Koppies deposit with identified mineralization in 6.4 sq km of palaeochannel. At the south of all these tenements is the Hirabeb area, comprising its largest tenement (730 km^2). In July 2020 the company announced prospectivity for a 36-km long palaeochannel here on the basis of geophysics (without any radiometric anomaly) and drilling five transects average 5.5 km apart. It is adjacent to Reptile's Aussinanis. Namib IV to the west has a 19 x 6 km palaeochannel.

In 2012 the company set about improving the grade of the beneficiation product. It now says that its *U-pgrade* process has wide potential for surficial deposits, especially calcrete ones, and it holds a provisional patent through its subsidiary Uranium Beneficiation Pty Ltd. This process became the primary focus of Marenica Energy. It has potential for Deep Yellow's Tumas ore and in Australia it was trialled for Toro Energy's Wiluna ore.

In May 2016 benefication results using Marenica Energy's U-pgrade process were reported for bulk samples from Deep Yellow's Tumas. Scrubbing produced good liberation of carnotite, then gravity separation gave process feed grade of 1.3% U_3O_8. Deep Yellow said that given its grades are much better than Marenica's own ores, Tumas would provide an excellent opportunity to commercialize the process. In September 2016 Deep Yellow and Marenica executed a technology licence agreement to use the U-pgrade process* for the Tumas project and other ores as it promised to dramatically reduce capital and operating expenses. However, this agreement was then suspended as Deep Yellow focused on other priorities for Tumas resources.

* The U-pgrade process takes run-of-mine calcrete ore at about 350 ppm. First is scrubbing and screening, then clay and calcite removal (about tenfold beneficiation to here), then carnotite concentration to give product of 10-15,000 ppm U_3O_8 (1 to 1.5%) with 85% of the original uranium. It can treat high-sulfate ore which is otherwise discarded. The product is suitable for acid leaching.

Elevate Uranium has a 75% interest in the project company, Marenica Minerals Pty Ltd, with Xanthos Mining Ltd holding 20% and Millennium Minerals 5%, both being Namibian. Elevate Uranium also has its fully-owned Mile 72 uranium prospect on the coast west of the main project leases.

Minor equity in Elevate Uranium Ltd is Hanlong Energy Ltd 18%, and Mingsun Technology Co Ltd 7%. China's Hanlong Energy Ltd is a subsidiary of privately-owned Sichuan Hanlong Group.

Zhonghe

Zhonghe Resources (Namibia) Development (Pty) Ltd is a Namibian registered company founded in 2008 by China Uranium Corporation Ltd (SinoU, now CNUC, 58%) a wholly-owned subsidiary of China National Nuclear Corporation (CNNC), and a private company, Namibia-China Mineral Resources Investment and Development (Pty) Ltd (Nam-China) (42%). It was looking at alaskites northeast of Swakopmund, close to Rössing with a view to open pit mining and heap leaching a low-grade (0.02%U) uranium deposit to produce about 600 tU/yr. Resources are believed to be 6000-12,000 tU. A mining licence was issued by the Ministry of Mines & Energy in November 2012, then the 2011 EIS was released in April 2013, this being the first public information on the project. In 2019 the company was continuing work to increase the resource base and planning leaching tests with Bannerman.

Other exploration

Xemplar Energy Corp of Canada was exploring its Cape Cross Uranium Project in the Namibian 'uranium corridor', near the coast.

Uranium One's Headspring Investments is investigating ISL potential in the Omaheke region.

Namibia's identified uranium resources are about 5% of the world's known total. Those recoverable at up to $130/kg are about 275,000 tonnes U. The Reasonably Assured Resources portion of this is 176,000 tU, accessible by open pit mining.

Electricity system

Namibia imports about 60% of its electricity, according to the African Union, relying on South Africa which faces its own supply constraints.

Inside Namibia are four main power plants owned and operated by Nampower: Ruacana (hydro, 330 MWe); Van-Eck (coal, 120 MWe); Paratus (diesel, 24 MWe); while the system is complemented by 22.5 MWe of emergency diesels at ANIXAS. Many homes and businesses generate their own power from petroleum, diesel or biomass but data on this is not effectively collected or compiled.

Namibia has set targets to raise domestic production to meet 80% of demand and connect 200,000 more homes to the grid.

Given Namibia's long experience of uranium mining, there is some interest in developing nuclear energy.

Organization

Mining is regulated under the Minerals (Prospecting and Mining) Act 1992, as well as the Atomic Energy Act 2005 and Environmental Management Act 2007. The Atomic Energy Board was established along with the National Radiation Protection Authority. Under the Minerals (Prospecting and Mining) Act 1992 the Minister must ensure that a 15% interest in any mineral licence must be held by Namibians.

Finland's Radiation & Nuclear Safety Authority (STUK) worked with Namibian authorities to help develop uranium mining policies and a safeguards and non-proliferation regime, under a program funded by the Finnish Foreign Ministry. As of early 2011 this did not include any development of a regulatory regime for nuclear power.

Non-proliferation

Namibia is party to the Nuclear Non-Proliferation Treaty and has had a comprehensive safeguards agreement in force since 1998 and in 2000 signed the Additional Protocol.

Namibia has also ratified the 1996 African Nuclear Weapon Free Zone Treaty, also known as the Pelindaba Treaty, which came into force in 2009 and precludes export of uranium to India.

Related information

Uranium Production by Country
World Uranium Mining Production